

Total No. of Questions : 8]

SEAT No. :

PD4625

[Total No. of Pages : 2

[6404]-131

B.E. (Electronics & Telecommunication Engineering)
MODERNIZED IOT
(2019 Pattern) (Semester - VII) (404184-E) (Elective - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Describe in detail the HTTP and MQTT session layer Protocol in IoT?[9]

b) Write a short note on IPv4 and IPv6 protocol. [8]

OR

Q2) a) Write a short Note on: [9]

i) Z Wave

ii) Bluetooth Low Energy

b) Compare TCP and UDP transport layer protocol? [8]

Q3) a) Differentiate between Raspberry Pi and Arduino Board. [9]

b) Explain the connection of microcontrollers with mobile devices through Bluetooth. [8]

OR

Q4) a) Explain the DHT interfacing deployment in ESP8266. [9]

b) List out and explain any 8 Arduino functions in detail. [8]

Q5) a) Draw and explain the Architecture of IIoT. [9]

b) Explain in detail about the Industrial Internet Architecture Framework (IIAF). [9]

OR

Q6) a) Briefly explain application of IIoT in: [9]

i) Healthcare

ii) Logistics

b) What is IIoT and How is it different from IoT? [9]

P.T.O.

- Q7)** a) Explain in detail the Air Pollution monitoring system. [9]
b) Describe the Sportsmen care application for vital signs monitoring in high-performance centers and fields. [9]

OR

- Q8)** a) Describe Photovoltaic Installation -Monitoring and optimization of performance in solar energy plants. [9]
b) Write a short note on: [9]
i) Smart Grid
ii) Smart Parking

